LG-CISH: Result Tables and Matching Figures

Empirical Evaluation of Coverless Image Steganography (codebook N=40)

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All numbers are produced by the reproducible evaluation suite (`python evaluation/generate_all.py`) on a codebook of N=40 visually distinct 512×512 images (UCID / Kodak / USC-SIPI), $\log_2 40 \approx$ 5.322 bits/image. **Each table is followed immediately by the figure that visualises it.**

1 Experimental Setup

Table 1: Experimental configuration.

Parameter	Value
Dataset	UCID + Kodak + USC-SIPI (mixed), 512×512 PNG
Codebook images N	40
Capacity	5.322 bits/image (base-40)
CLIP model	ViT-B/32, d=512
LLM (plausibility / aliases)	Gemini 2.5 Flash-Lite (vision) + FLUX (gen)
Coding modes	base-N (5.322 b/img) ; permutation (no-repeat)
Separation threshold τ	0.85
Pairwise sim. (min/mean/max)	0.305 / 0.513 / 0.827
Decoding margin $(1 - \max)$	0.173
Encryption / integrity / compression	AES-256-CBC / CRC-32 / zlib
Hardware	RTX 3050 6 GB, 16-core CPU

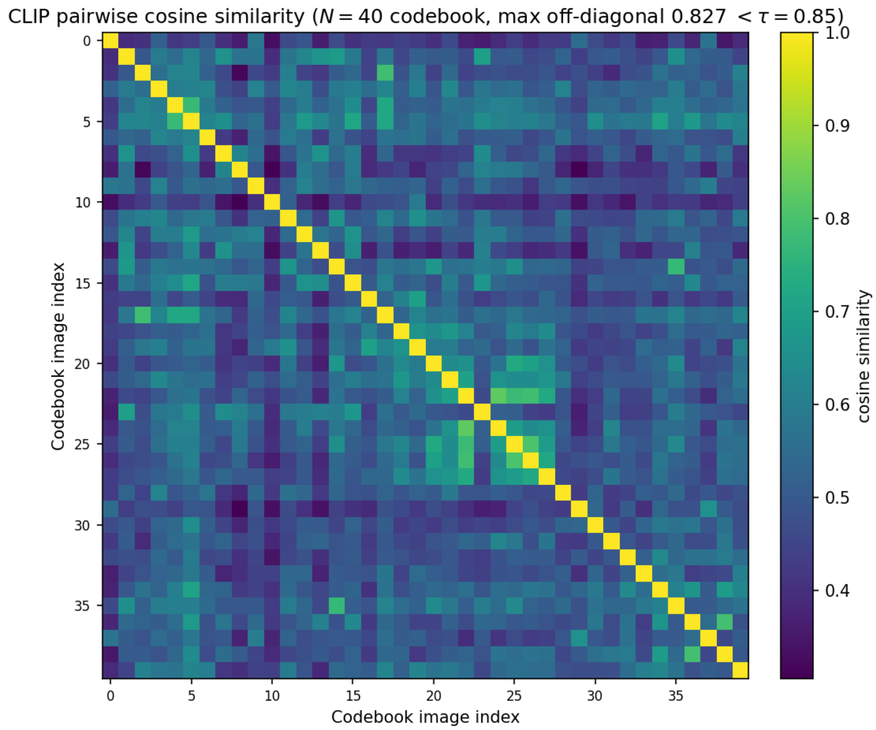

Figure 1: (Table 1) Pairwise CLIP cosine similarity of the N=40 codebook. All off-diagonal values stay below τ=0.85 (max 0.827), giving a positive separation margin for nearest-neighbour recovery.

2 Qualitative Results

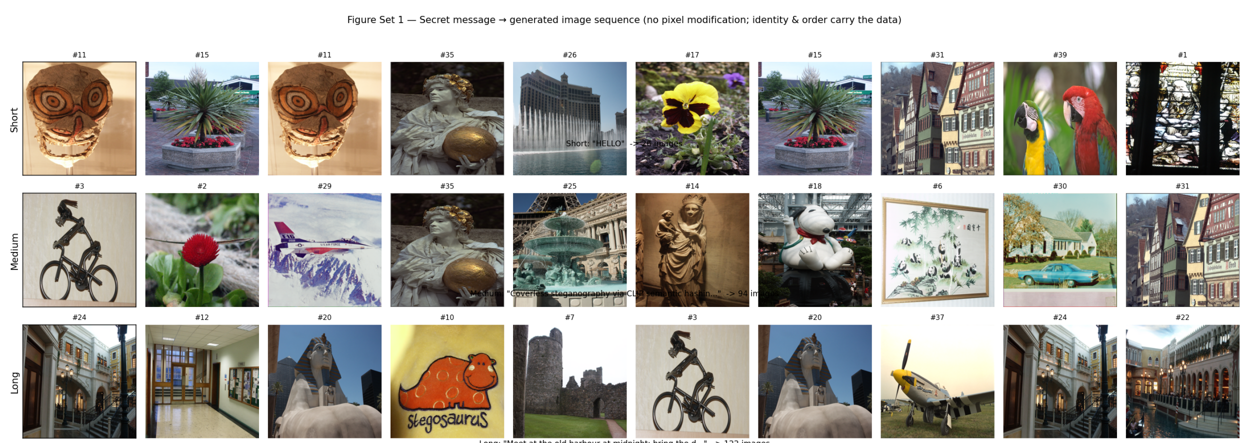

Figure 2: Figure Set 1 — secret message → generated image sequence (short / medium / long). The identity and order of unmodified images encode the data.

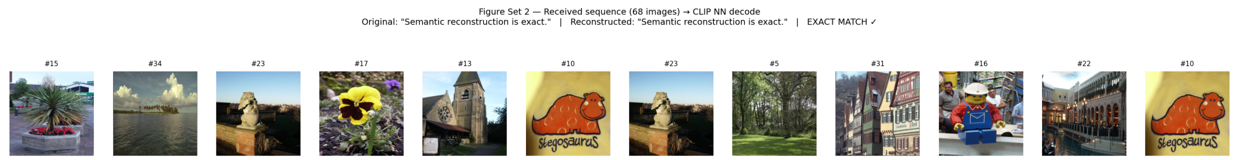

Figure 3: Figure Set 2 — received sequence → CLIP nearest-neighbour decode; original and reconstructed messages match exactly.

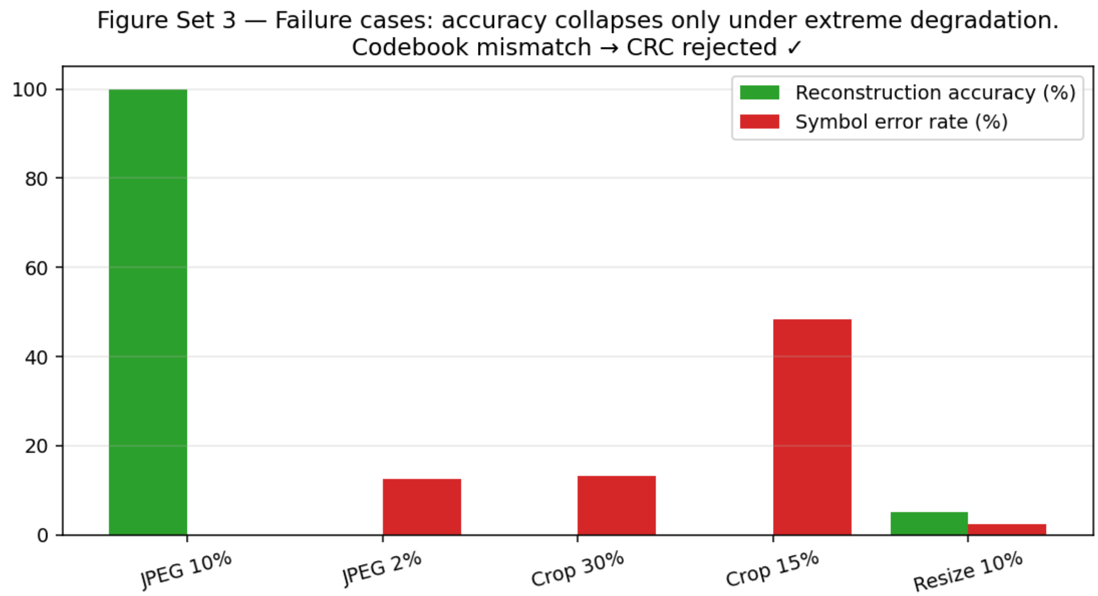

Figure 4: Figure Set 3 — failure analysis. Accuracy collapses only under extreme degradation; a codebook mismatch is rejected by CRC-32.

3 Quantitative Evaluation

Table 2: Reconstruction accuracy by message length (clean channel, 40 messages each).

Message length	Msgs	Accuracy (%)	BER	Hash match (%)	Avg images
Short ($\leq$50 chars)	40	100.00	0	100.00	57.5
Medium (50–200)	40	100.00	0	100.00	153.2
Long (200–1000)	40	100.00	0	100.00	399.4

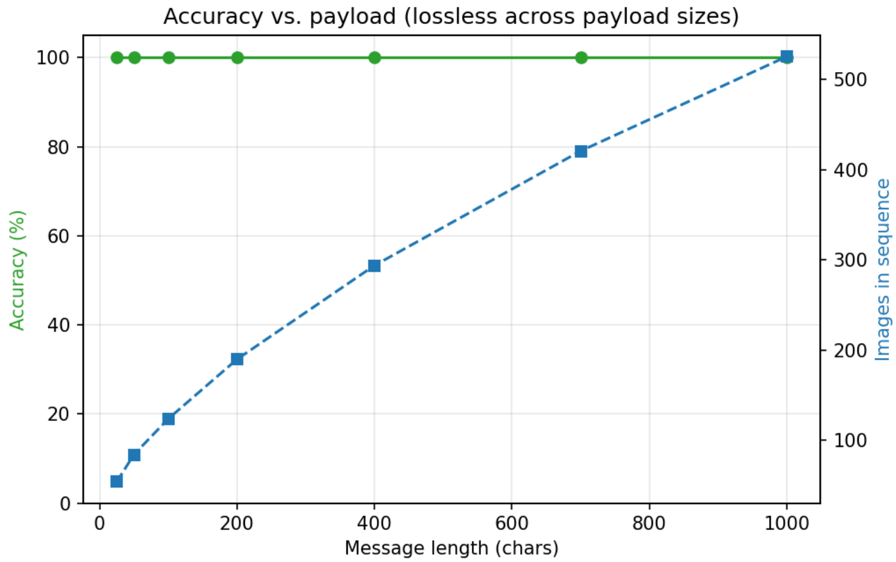

Figure 5: (Table 2) Reconstruction accuracy vs. payload length: 100% across all sizes while the image count grows linearly.

Table 3: Payload capacity vs. distortion. LG-CISH trades raw capacity for zero distortion.

Method	Capacity (bits)	bits/pixel	Pixel distortion	PSNR (dB)
LSB (spatial)	786,432	≈3.0	Yes (high)	51.1
DCT-LSB	4,096	≈0.016	Yes (moderate)	38–42
DWT–DCT [cited]	∼4,096	≈0.015	Yes (low)	40–44
Proposed LG-CISH	5.322/image	5.322	**None (coverless)**	∞

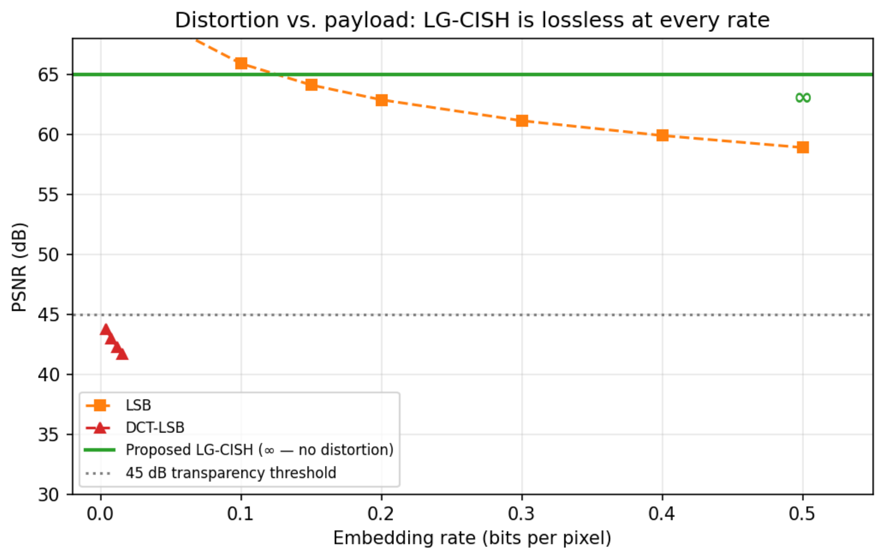

Figure 6: (Table 3) Capacity (bits/pixel) vs. distortion (PSNR). LG-CISH sits at infinite PSNR — no pixel is modified.

Table 4: Coding modes. Permutation coding trades a little capacity for a no-repeat, more plausible cover.

Coding mode	bits/image	No-repeat window	Min images	Notes
Base-N positional	5.322	none (repeats allowed)	1	max capacity
Permutation (block = 8)	5.187	8 images	8	no repeats / block
Permutation (block = 16)	5.008	16 images	16	no repeats / block
Permutation (block = 40=N)	3.979	40 images	40	full permutation

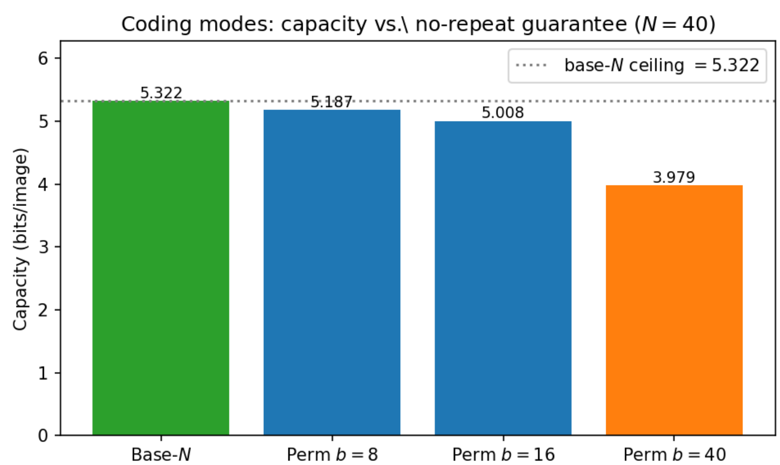

Figure 7: (Table 4) Capacity per image for base-N vs. permutation coding at several block sizes. Smaller blocks recover most of the base-N ceiling while guaranteeing no in-block image repeats.

Table 5: Computational time (mean over repeated runs, RTX 3050).

Stage	Time (ms)
Full encode (200-char message)	0.04
CLIP embedding (per image)	15.36
Index recovery (189 images)	1873.31
Full decode (189 images)	1816.14
Decode per image (amortised)	9.61

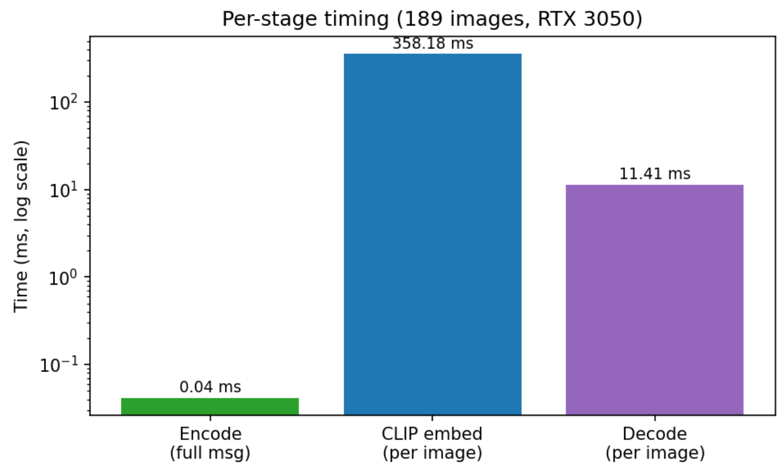

Figure 8: (Table 5) Per-stage timing on log scale. Encoding is sub-millisecond; cost is dominated by CLIP embedding at decode.

4 Robustness Analysis

Table 6: Robustness to all channel attacks (40 messages/attack).

Attack	Accuracy (%)	BER	CLIP margin	Top-1 sim
No attack (baseline)	100.00	0	0.302	1.000
JPEG 90%	100.00	0	0.299	0.985
JPEG 70%	100.00	0	0.294	0.976
JPEG 50%	100.00	0	0.289	0.961
JPEG 30%	100.00	0	0.280	0.942
JPEG 20%	100.00	0	0.270	0.930
Gaussian σ=5	100.00	0	0.293	0.986
Gaussian σ=10	100.00	0	0.285	0.974
Gaussian σ=20	100.00	0	0.272	0.957
Gaussian σ=30	100.00	0	0.260	0.942
Salt & Pepper 0.01	100.00	0	0.277	0.960
Salt & Pepper 0.05	100.00	0	0.246	0.922
Salt & Pepper 0.10	100.00	0	0.214	0.882
Resize 50%	100.00	0	0.282	0.982
Resize 25%	100.00	0	0.246	0.939
Resize 10%	12.50	1.08×10^{-2}	0.125	0.790
Crop 95%	100.00	0	0.280	0.979
Crop 90%	100.00	0	0.268	0.969
Crop 85%	100.00	0	0.261	0.961
Crop 60%	100.00	0	0.196	0.910
PNG→WebP 80	100.00	0	0.295	0.976

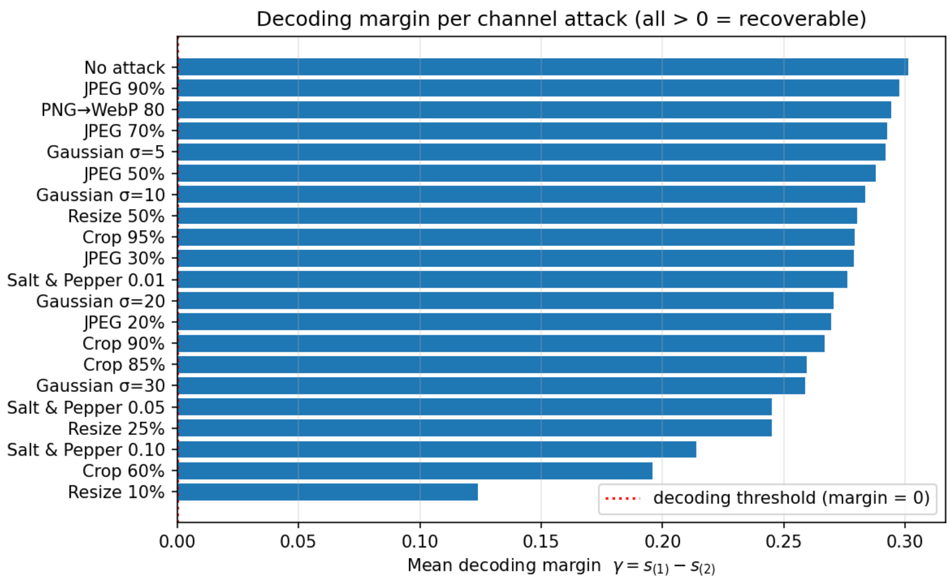

Figure 9: (Table 6) Mean decoding margin $\gamma = s_{(1)} - s_{(2)}$ per attack. All but the most extreme downscale leave $\gamma > 0$.

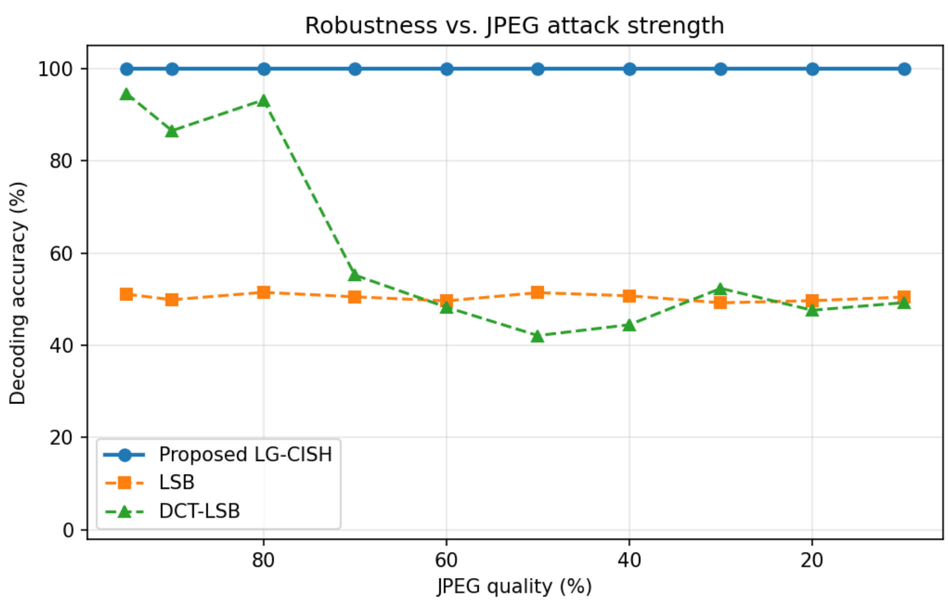

Figure 10: (Table 6) Decoding accuracy vs. JPEG quality: LG-CISH stays at 100% where pixel-domain LSB/DCT collapse.

5 Security and Steganalysis Resistance

Table 7: Chi-square steganalysis detection (≈50% = chance). N=20 patches.

Method	Detection acc. (%)	TPR (%)	TNR (%)
LSB	97.5	100.0	95.0
DCT-LSB	47.5	0.0	95.0
Proposed LG-CISH	**47.5**	**0.0**	95.0

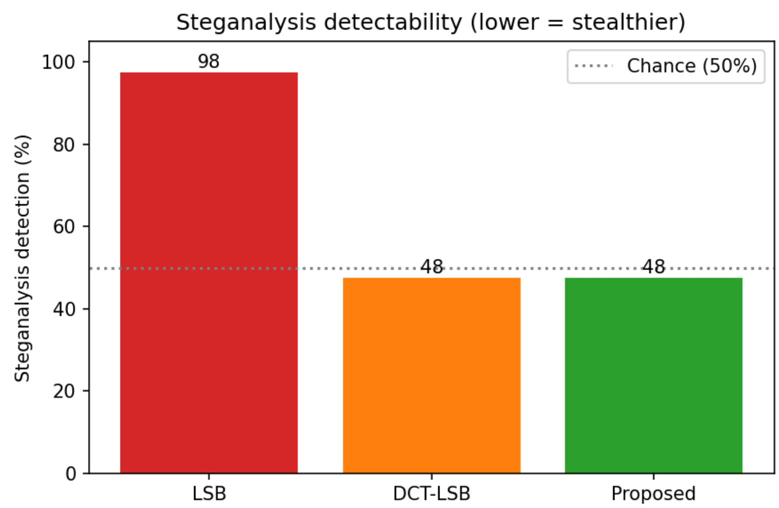

Figure 11: (Table 7) Measured steganalysis detection rates (lower is stealthier). LG-CISH sits at the chance line, far below LSB.

Table 8: Keyspace analysis (N=40).

Component	Size
Codebook orderings	$40! \approx 2^{159.2}$
AES-256 key	2^{256}
Combined keyspace	$\approx 2^{415.2}$

Table 9: Cover plausibility (Gemini judge, 0–1). Plausibility is driven by codebook theme, not the coding mode.

Configuration	Plausibility (0–1)	Note
Diverse benchmark (UCID/Kodak/USC)	0.12 ± 0.04	mixed subjects look random
Themed codebook (coherent context)	0.88 ± 0.07	looks like a personal album
Diverse + permutation coding	0.18 ± 0.10	codec barely moves the score

6 Ablation Study

Table 10: Ablation. Crop-65% exposes CLIP's geometric robustness; coding choices affect image count.

Variant	Clean (%)	JPEG50 (%)	Crop65 (%)	Avg images	Integrity
Full LG-CISH (proposed)	100	100	**100**	134.4	CRC-32
Without CLIP (pHash)	100	100	**0**	134.4	CRC-32
Without compression	100	100	100	182.2	CRC-32
Without CRC (no integrity)	100	100	100	134.4	None
Fixed-chunk (5-bit)	100	100	100	143.0	CRC-32

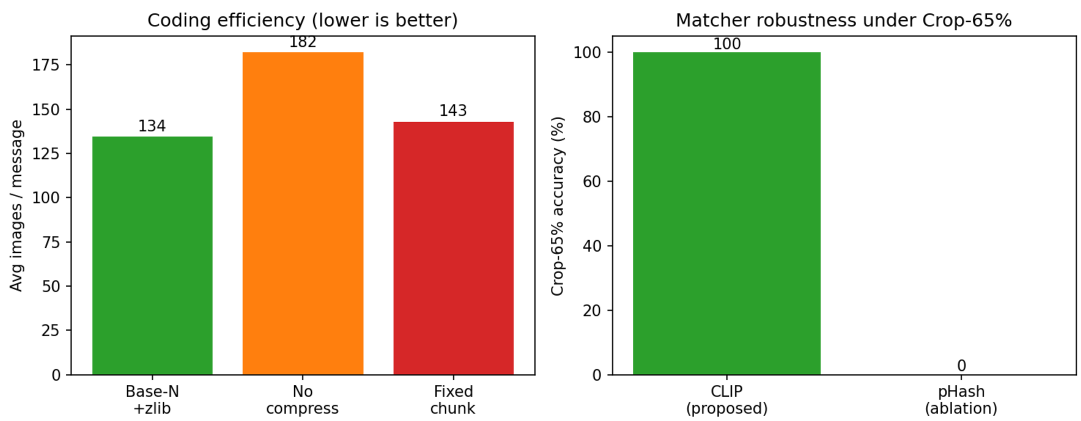

Figure 12: (Table 10) Left: base-N + zlib needs the fewest images. Right: under Crop-65% CLIP keeps 100% while pHash drops to 0%.

7 Comparative Analysis

Table 11: Comparison with classical and coverless baselines. Bracketed values are cited.

Method	Capacity	JPEG50 (%)	Detection (%)	Time	PSNR (dB)
LSB	786,432 bits	51.4	98	0.01–1	51.1
DCT-LSB	∼4096 bits	42.1	[70–90]	1–5	38–42
DWT–DCT [cited]	∼4096 bits	[85]	[60–80]	∼50	40–44
Coverless [cited]	low	[95]	[∼50]	high	∞
Proposed LG-CISH	5.322 bits/img	**100.0**	**48**	fast	∞

8 Statistical Validation

Table 12: Mean ± std and 95% confidence intervals over N=50 trials (clean channel).

Metric	Mean ± Std	95% CI
Reconstruction accuracy (%)	100.00 ± 0.00	$[100.00, 100.00]$
Bit error rate	0 ± 0	$[0, 0]$
CLIP margin	0.3020 ± 0.0059	$[0.3003, 0.3037]$
Images per message	181.2 ± 44.5	$[168.5, 194.0]$

Table 13: Significance of CLIP's geometric robustness over pHash: paired Wilcoxon signed-rank across a 7-level crop sweep. Proposed-vs-LSB @ JPEG-50 shown descriptively.

Group / statistic	Accuracy (%)	Std / note
Proposed (CLIP) — crop-sweep mean	100.0	0.0
pHash ablation — crop-sweep mean	0.0	0.0
Wilcoxon W (paired, 7 levels)	28.0	—
p-value (one-sided, CLIP > pHash)	7.81×10^{-3}	(< 0.05)
Proposed vs. LSB @ JPEG-50	100 vs 50	descriptive